YouTube Trending Video Analytics

1. Introduction
YouTube trending videos provide useful information about content popularity and audience engagement. This project analyzes trending video records across India, the USA, and Great Britain to identify category popularity, sentiment patterns, and regional differences.

2. Abstract
The project uses Python, SQL, and data visualization to study YouTube trending video datasets. The data is cleaned and standardized, titles and tags are classified by estimated sentiment, categories are ranked by average views, and regional and time-based patterns are visualized.

3. Tools Used
Google Colab, Python, Pandas, NumPy, Matplotlib, Seaborn, TextBlob, SQLite, SQL, and ReportLab. Tableau can be used to create an additional interactive dashboard.

4. Steps Involved in Building the Project

Step	Description
Data collection	Downloaded country-wise video CSV and category JSON files from Kaggle.
Data cleaning	Handled missing values, standardized fields, and removed duplicate records.
Sentiment analysis	Classified video titles and tags as positive, neutral, or negative.
SQL analysis	Ranked categories by average views and compared regional engagement.
Visualization	Created category, sentiment, regional, and time-series charts.
Reporting	Exported the analysis results and generated the final PDF report.

5. Conclusion
The cleaned dataset contains 112,735 records from 3 regions. SQL queries and visualizations provide a way to compare category popularity, estimated sentiment, regional engagement, and trending activity over time. Findings depend on the available dataset and its limitations.